

Cluster Genetic of rat's liver to different chemicals

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# First, in R download and install Bioconductor using

if (!require("BiocManager", quietly = TRUE))
  install.packages("BiocManager")
BiocManager::install(version = "3.19")

# Then install the GEOquery package:

BiocManager::install("GEOquery")

# Once installed, load this package
library(GEOquery)

drugdata <- getGEO("GSE57815")[[1]]
#methods(class=class(drugdata))
# Aside: The getGEO function will download a list - usually there is only one component in the list as

help("ExpressionSet-class")

# The expression data can be extracted using the exprs() extractor function:
expression=exprs(drugdata) # this is G by N (number of genes by number of samples)

# The phenotypic data can be extracted using the pData() extractor function:
pData(drugdata) # this is N by P (number of samples by number of phenotypic variables)

# select samples : Dose = 0 mg/kg, any duration
baseline2 <- drugdata[, pData(drugdata)$"dose:ch1" == "0 mg/kg"]
x = exprs(baseline2)

#install.packages("matrixStats")
library(matrixStats)

xt = t(x) # transpose of data
sds = colSds(xt) # standard deviations of each rows and columns
boxplot(sds)

# filtering variance > median variance across genes
xt_filter = xt[, sds > median(sds)]
dim(xt_filter)

# Similarity measure: use absolute correlation between genes
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cormat = cor(xt_filter)
dim(d)

# distance
d = 1 - abs(cormat)

# Hierarchical clustering
hclust_complete = hclust(as.dist(d), method = "complete")
plot(hclust_complete, labels = FALSE)
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